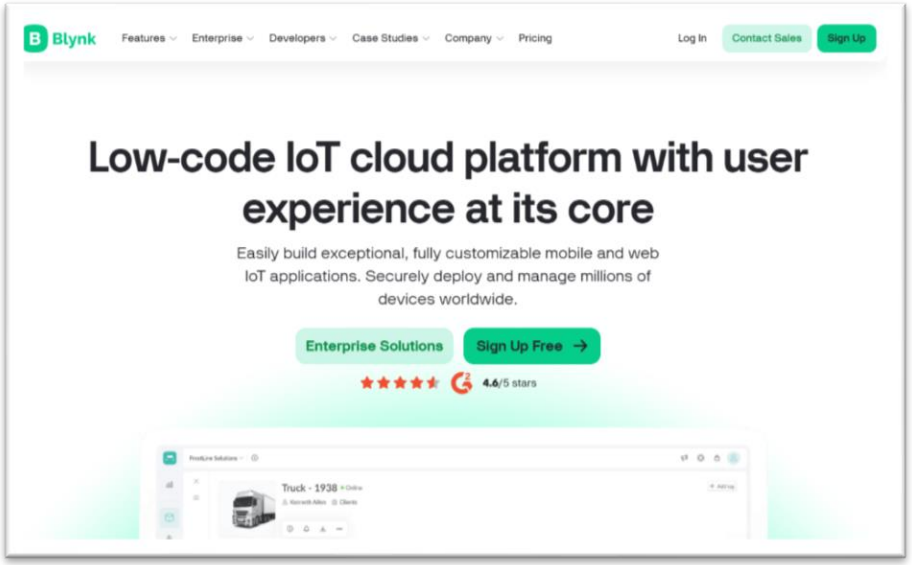
ICT30016 IoT Track – Week 5

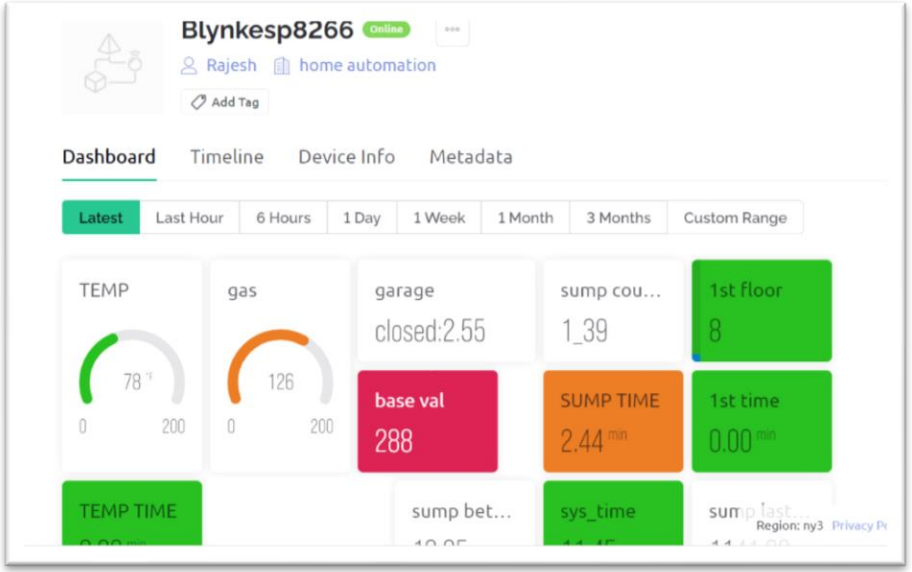
Using Blynk with NodeMCU ESP32

Prepared by Dr. Mark Tee Kit Tsun (SUTS)

Why Blynk ?

- Blynk is an Online Platform that provides a convenient Web Dashboard for your ESP32-based IoT machine.
- It has both web and mobile app interfaces so you can display buttons, labels, graphs and other widgets for displaying data from your ESP32, or trigger functions to execute.
- It does not have any direct option for interacting with other external services (e.g. database, virtual machines, AI, social media, etc.).
- Although Blynk has some limited machine-to-machine communication options (Automation), they require paid subscription to unlock.
- Still, it is useful for quickly testing your IoT system and demonstrating its capabilities.
- To know more, visit the official website: https://blynk.io/



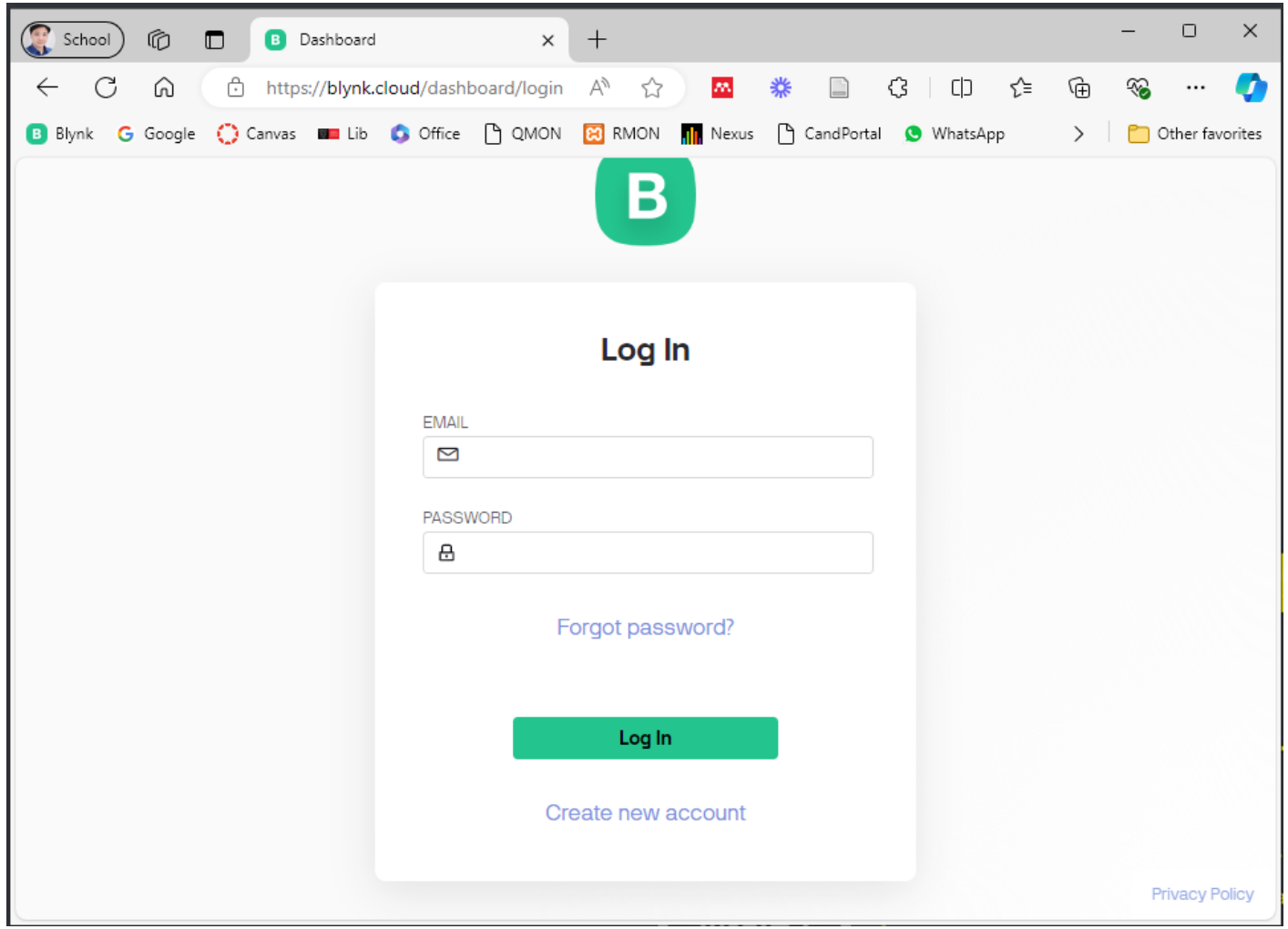


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First, Create Your Free Account (If you don't have one)

- Go to https://blynk.cloud/dashboard/login
- Create a New Account and Activate it using your Email.
- Log into Blynk.
- You will see your Blynk.Console

The Blynk.Console is the hub where you manage your projects (Developer Zone), connected Devices, their Users and even your Organisation/Location details.

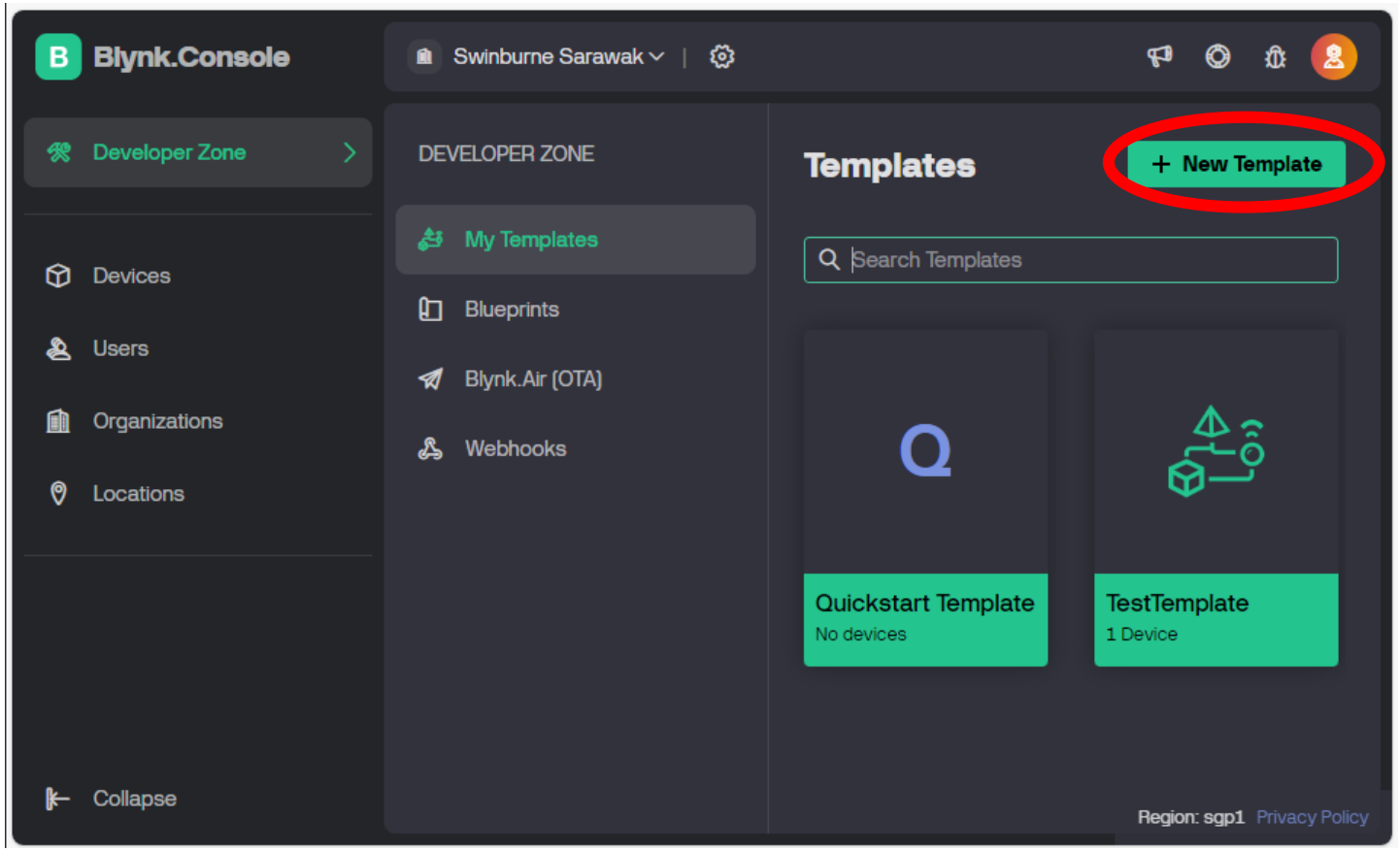


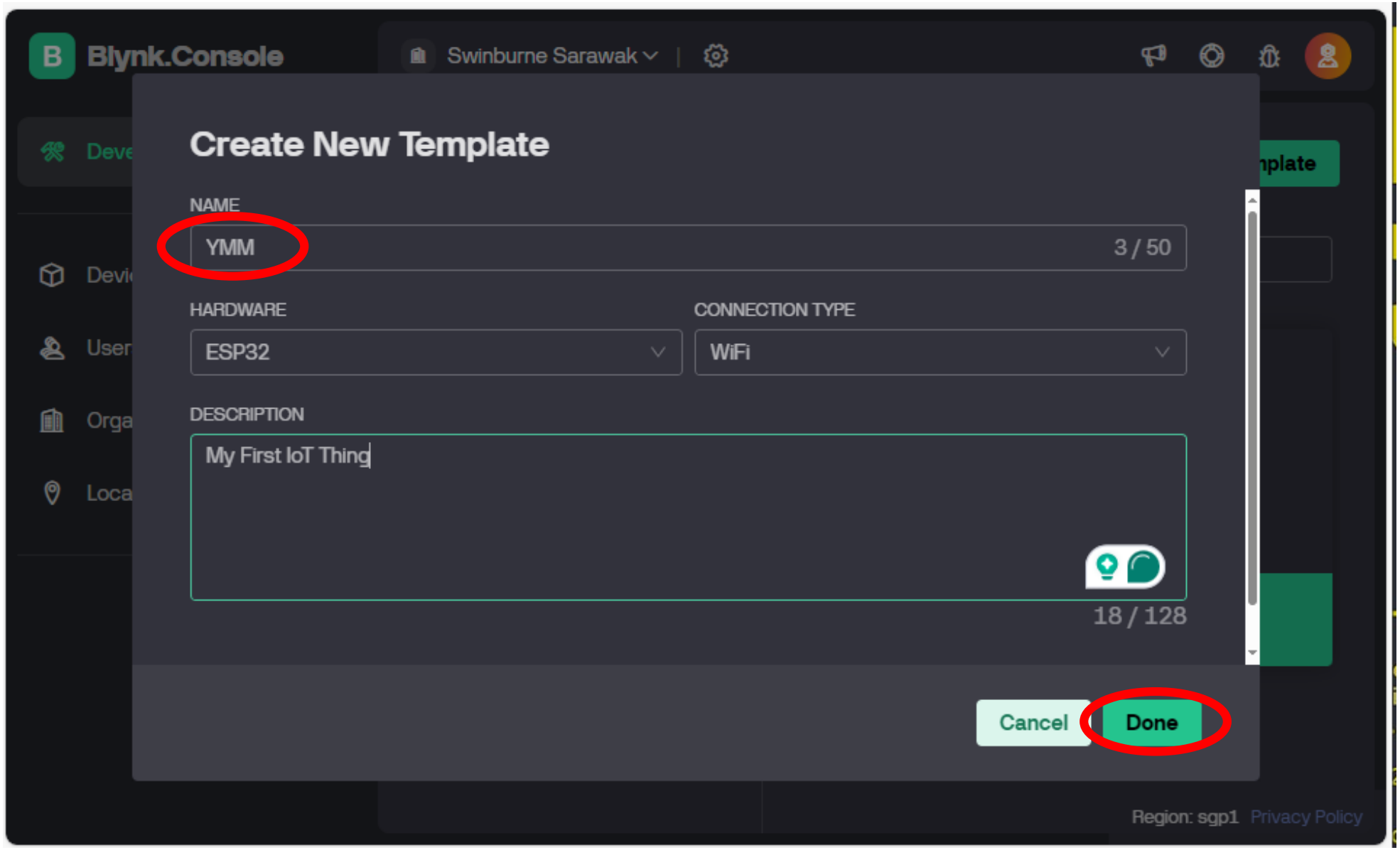
Create Your 1st Template

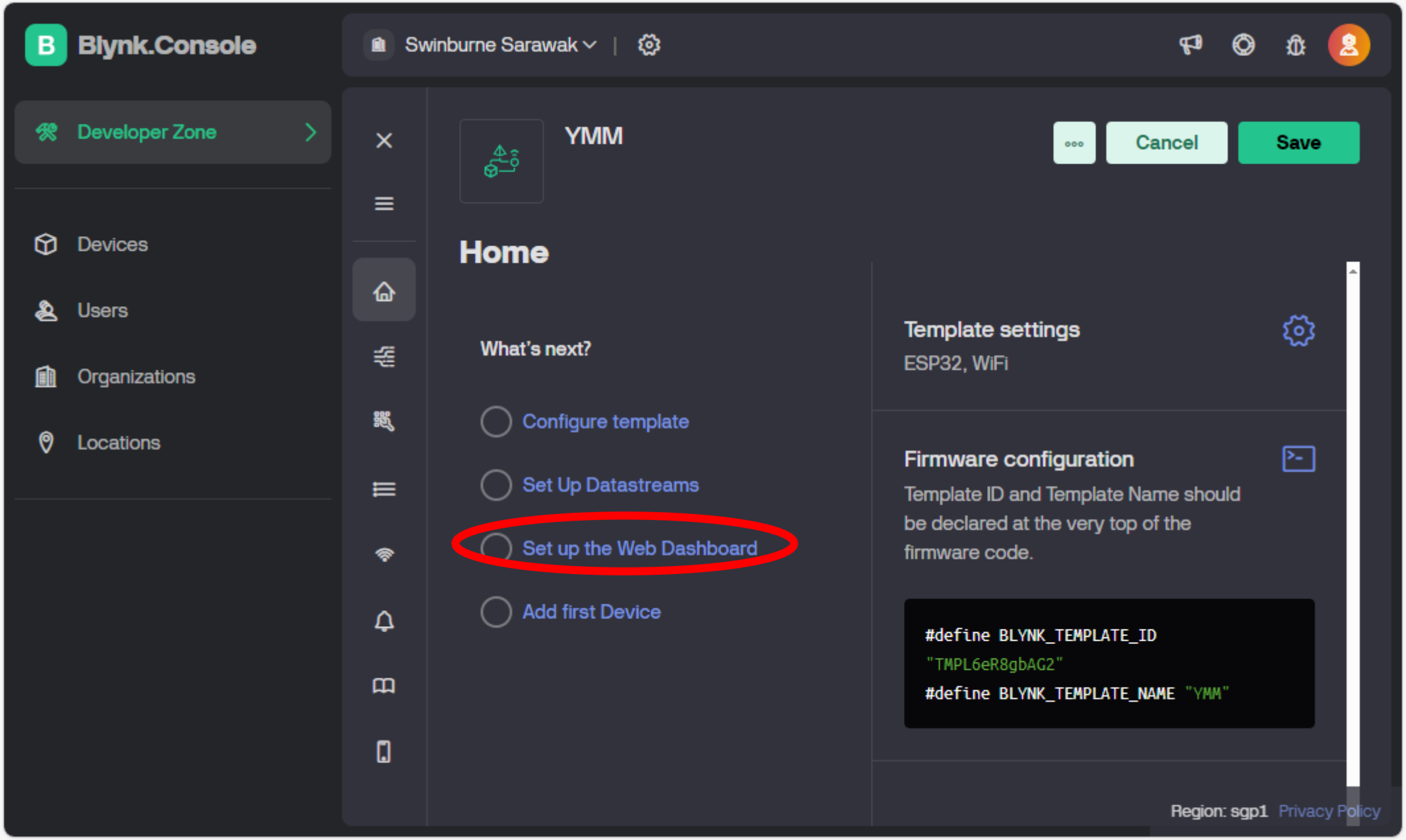
- Go to “Developer Zone” and Choose “+ New Template”.

A Template is an IoT Project. It holds information such as the Datastreams, Dashboard Layout and other project information.

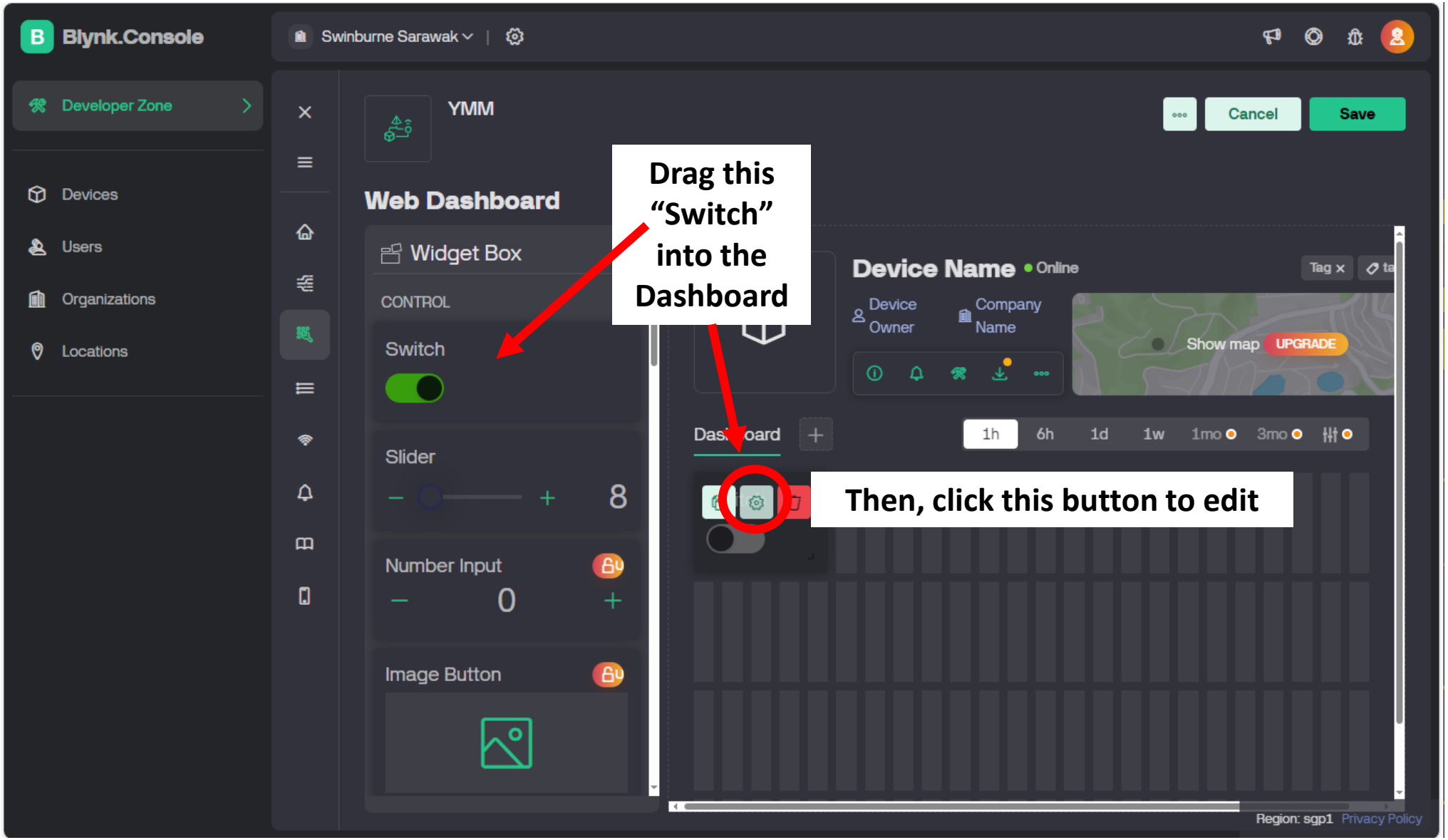
Once a Template is built, you can add multiple ESP32 devices that can be controlled using the Template’s contents.



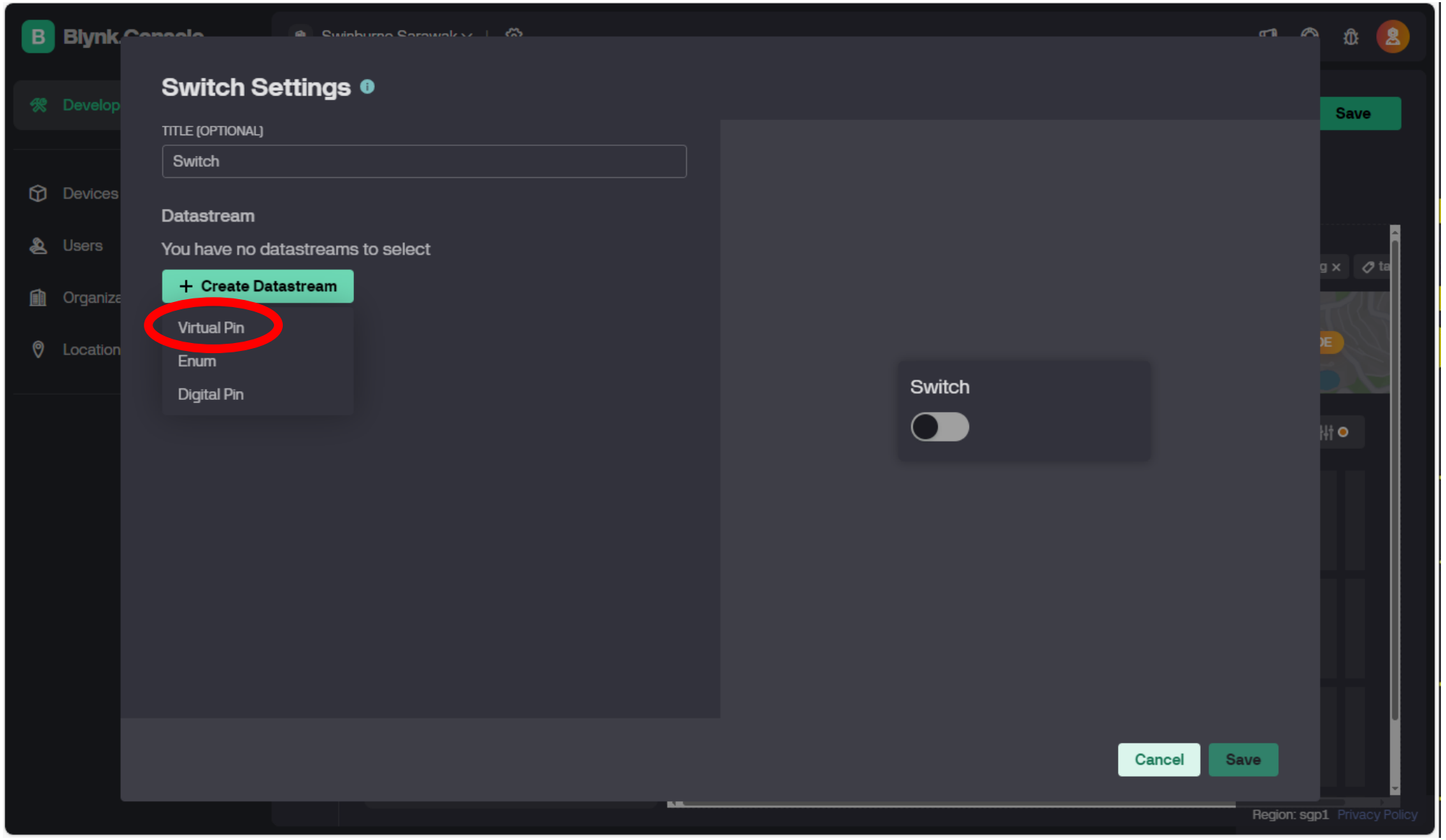

Blynk.Console
Swinburne Sarawak
Create New Template
NAME
YMM
3 / 50
HARDWARE
ESP32
CONNECTION TYPE
WiFi
DESCRIPTION
My First IoT Thing
18 / 128
Cancel
Done
Region: sgp1
Privacy Policy


Blynk.Console
Swinburne Sarawak
Developer Zone
Devices
Users
Organizations
Locations
YMM
Cancel
Save
Home
What's next?
Configure template
Set Up Datastreams
Set up the Web Dashboard
Add first Device
Template settings
ESP32, WiFi
Firmware configuration
Template ID and Template Name should be declared at the very top of the firmware code.
#define BLYNK_TEMPLATE_ID "TMPL6eR8gbAG2"
#define BLYNK_TEMPLATE_NAME "YMM"
Region: sgp1
Privacy Policy

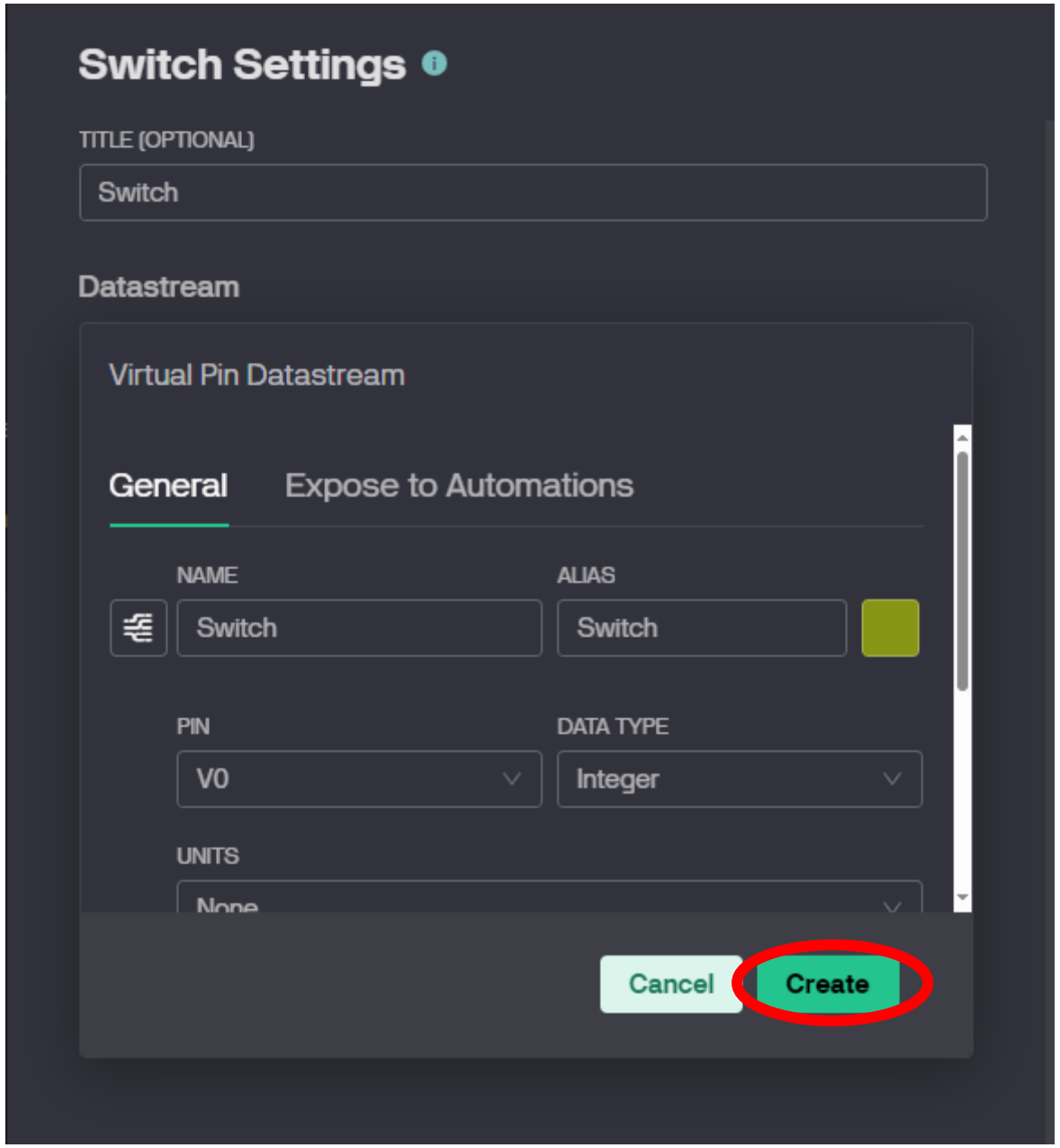
- Here, you can choose the type of widget you need, drag and drop it into the Dashboard, then click the Settings button to configure it.



- Choose to attach this control widget to a new Datastream, and select “Virtual Pin”.
- We use Virtual Pins to interact between the Dashboard and your ESP32.

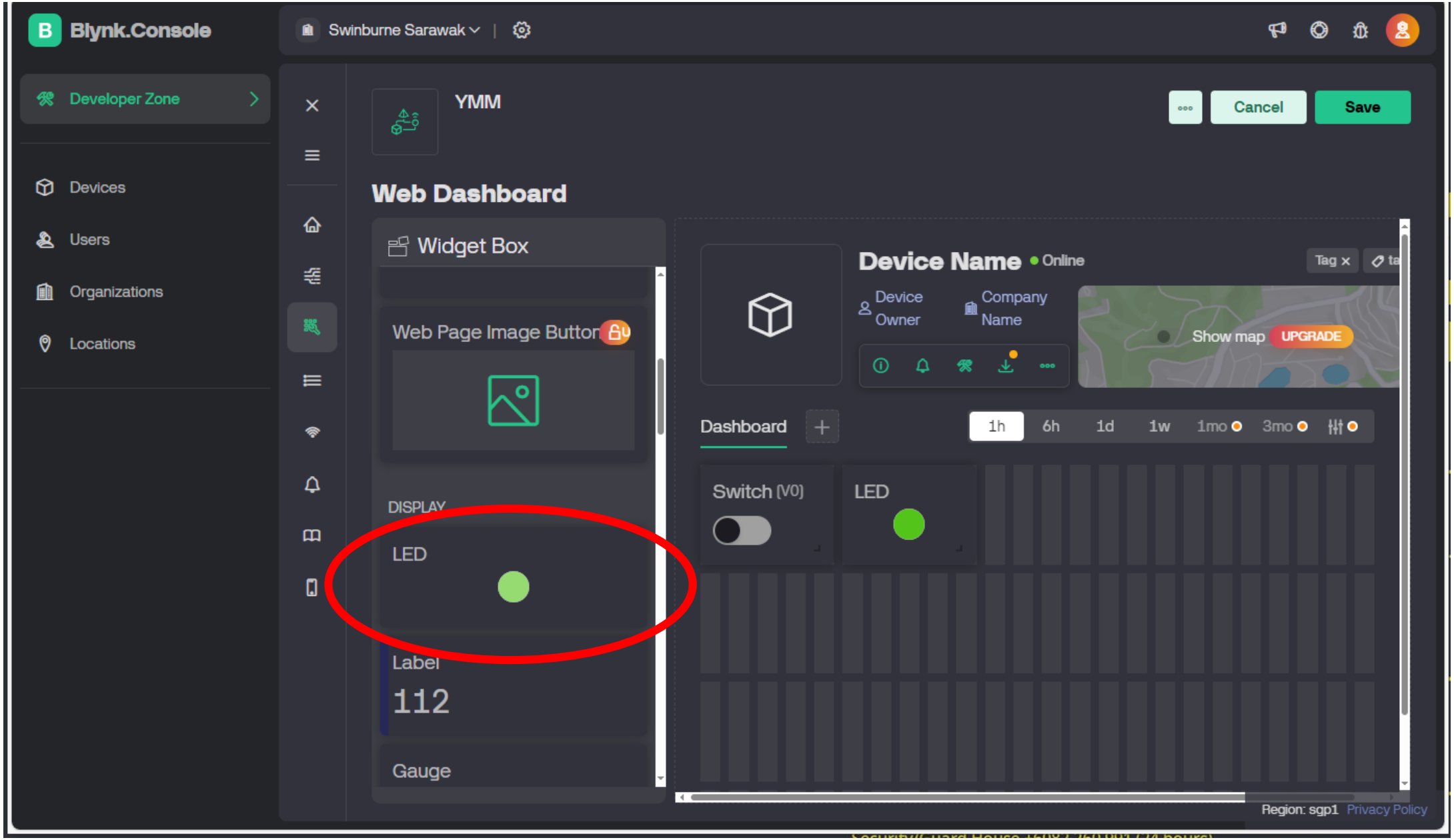


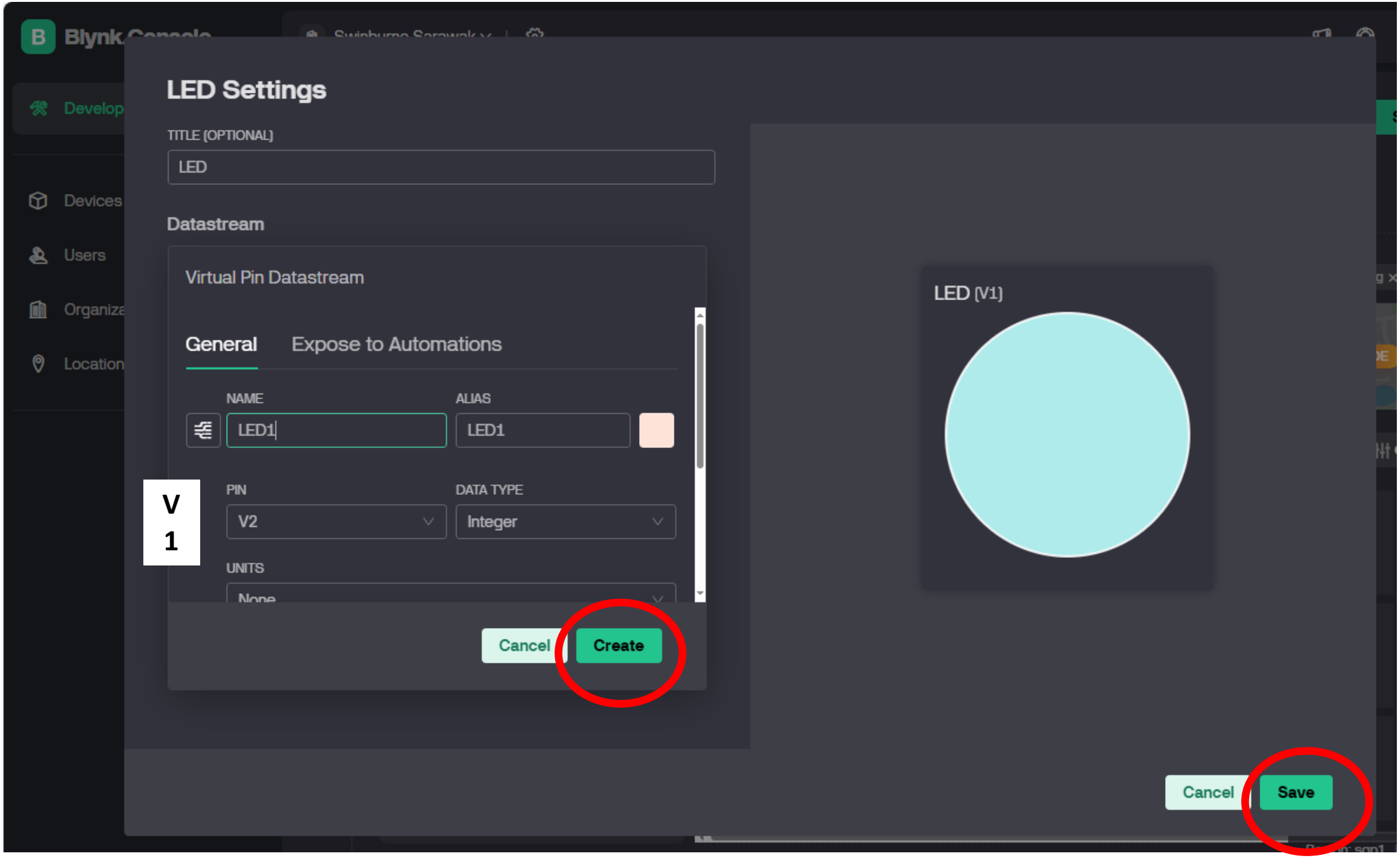
- You can configure details in the Virtual Pin.
- Pay close attention to which Pin this is (V0, V1, V2, V3, or V4).
- The Free account limits the choice to only 5 Datastreams.
- When done, press “Create”.
- You will now return to the Dashboard Editor to add other widgets.



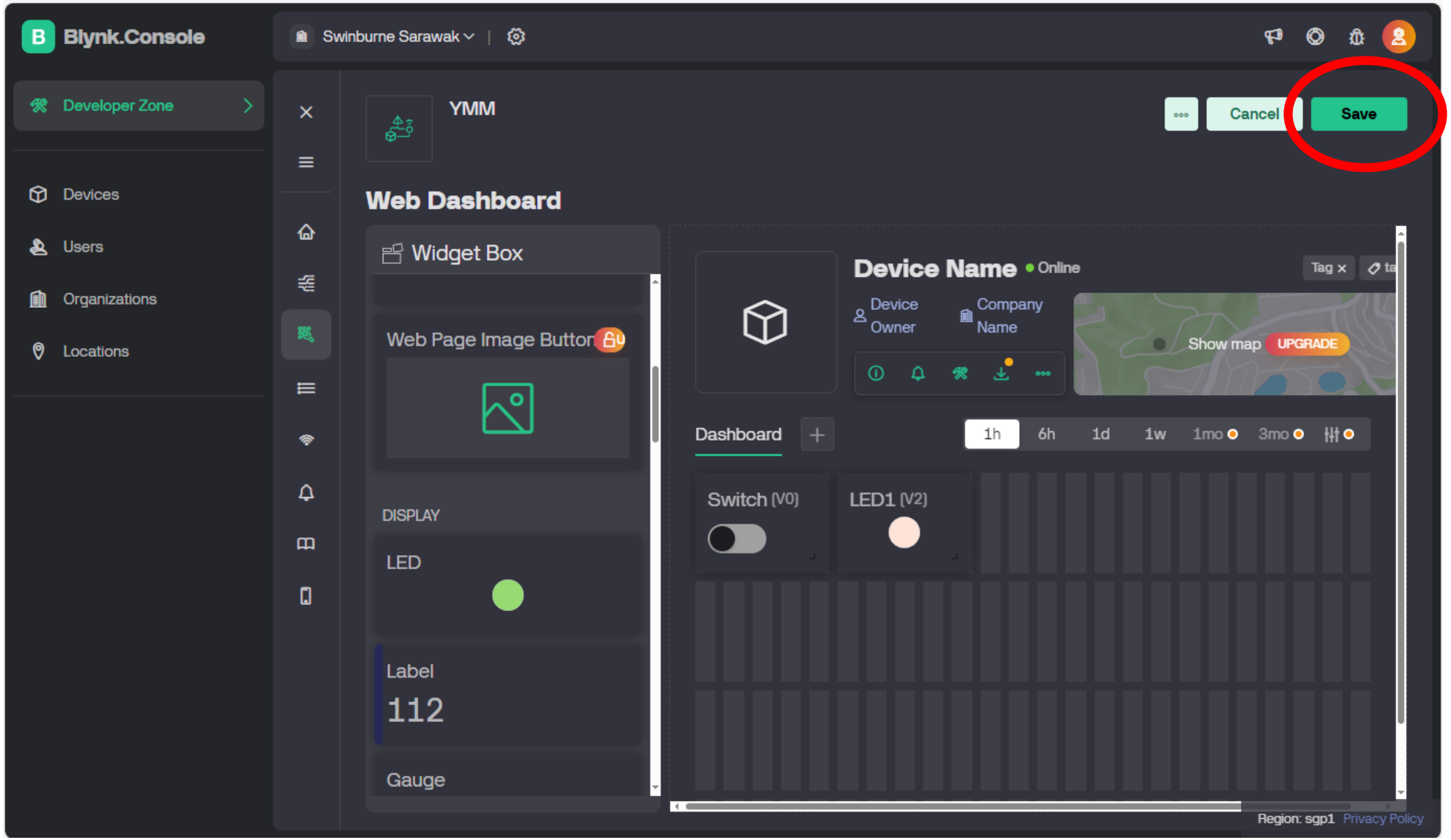
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- Adding another Widget now to represent Digital Output (LED) on the Dashboard.

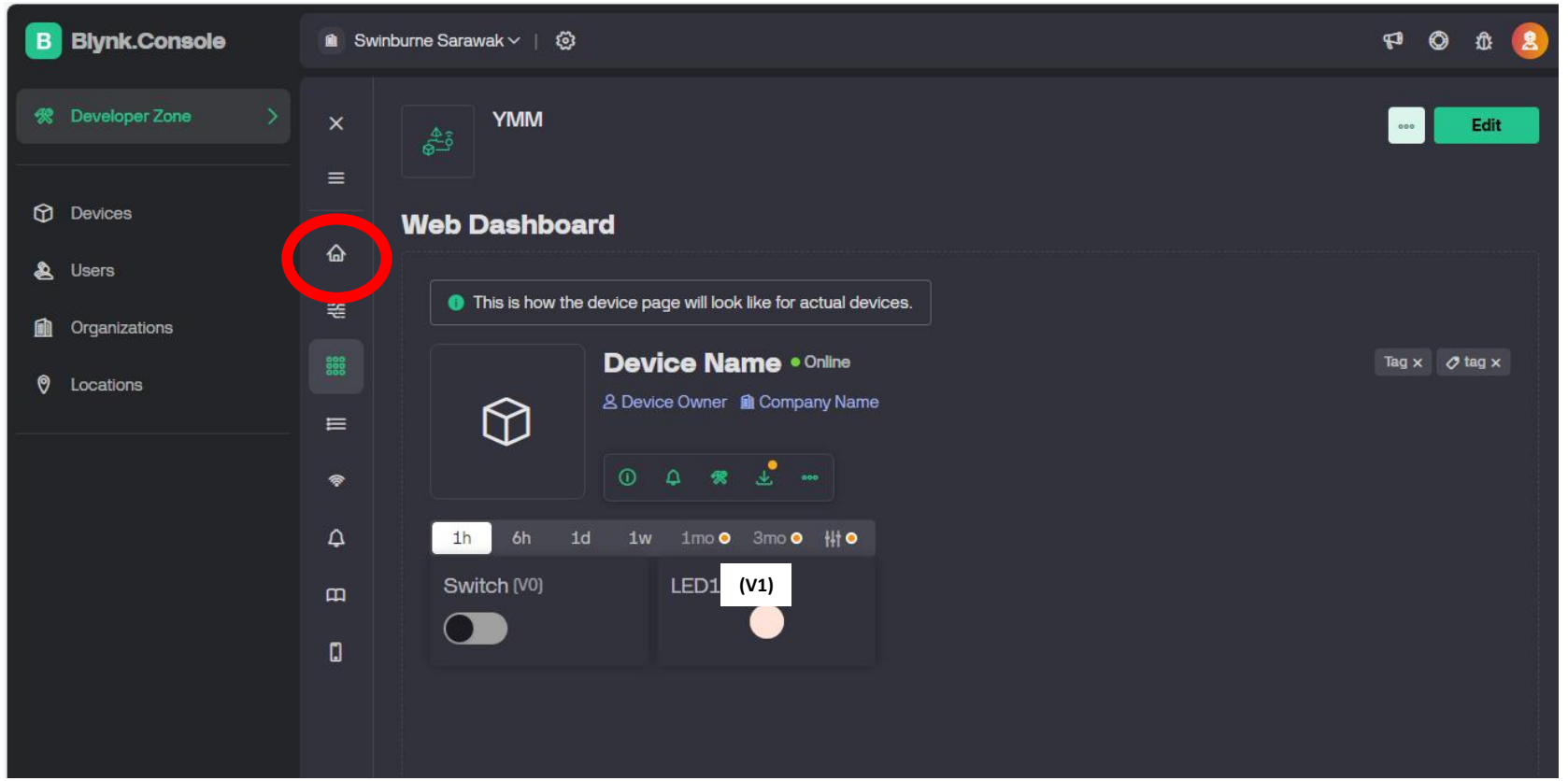



LED Settings
TITLE (OPTIONAL)
LED
Datastream
Virtual Pin Datastream
General
Expose to Automations
NAME
ALIAS
LED1
LED1
PIN
DATA TYPE
V2
Integer
UNITS
None
Cancel
Create
LED (V1)
Cancel
Save
V
1

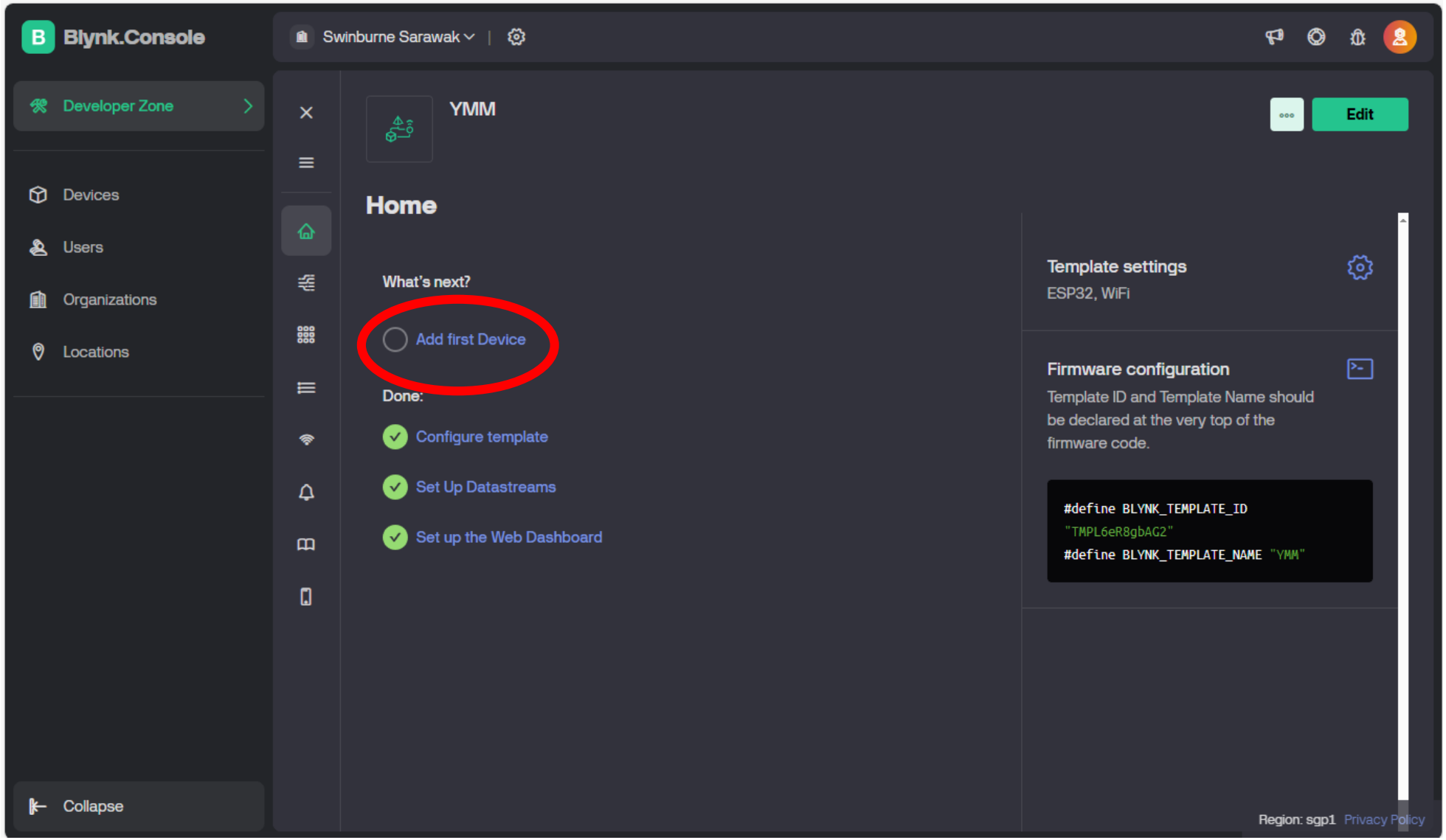
- After you are done, click “Save” to save work on the Dashboard into the Template.



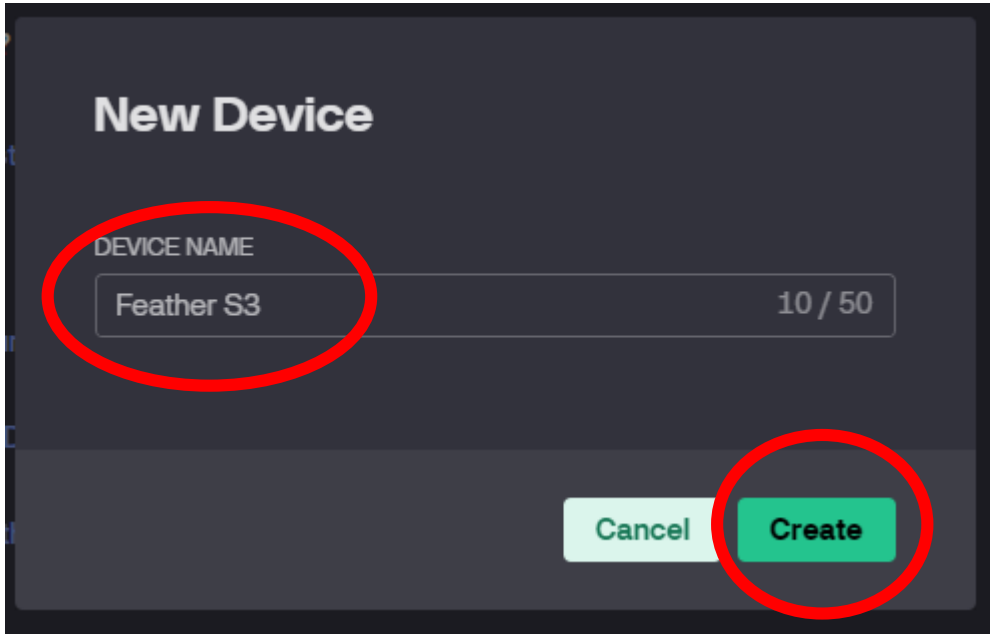
- If you have followed the previous steps, you have added 2 Control Widgets:
  A Switch (Virtual Pin V0), and a LED (Virtual Pin V1).
- Now, click this button to return to the Template Home Page.



- At the Template Home page, click “Add first Device”.
- Remember, it is possible for multiple ESP32 machines to be controlled through separate Blynk Dashboards.

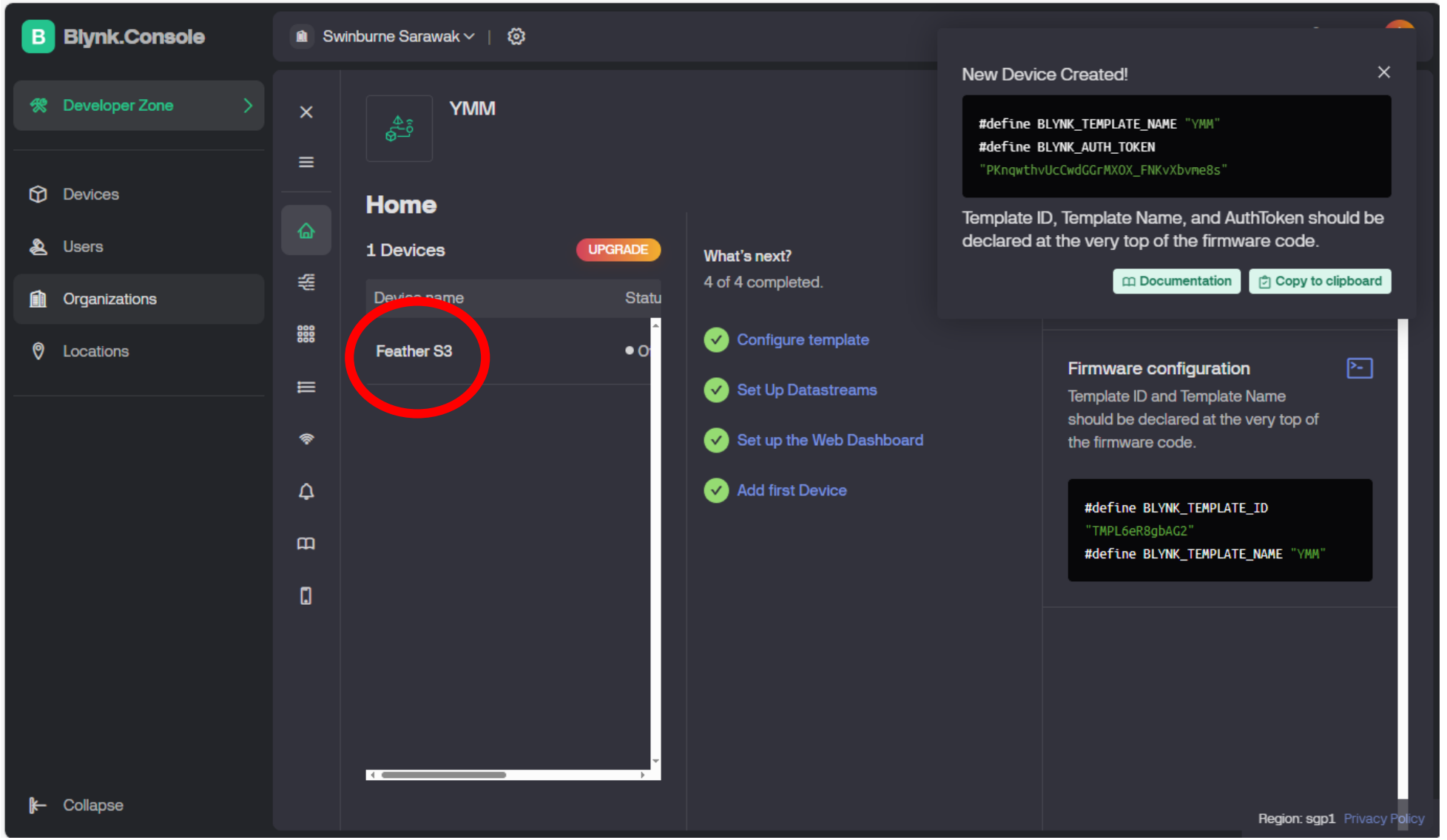


- This step is very important. Make sure to apply a unique name to identify your ESP32 machine.
- Once configured, this Blynk Device's Dashboard will be linked to your ESP32 board using a few lines of Identification Code.

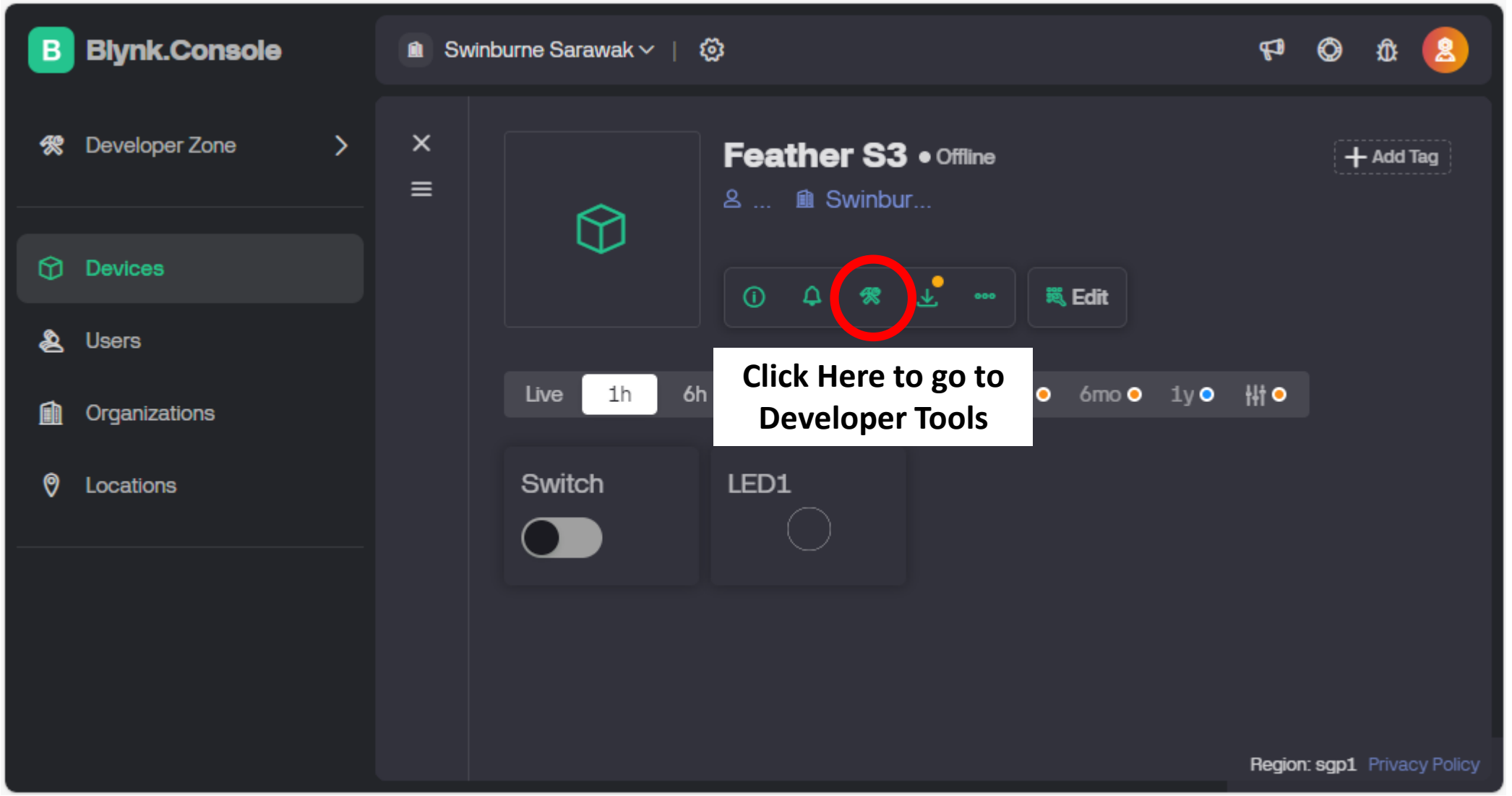


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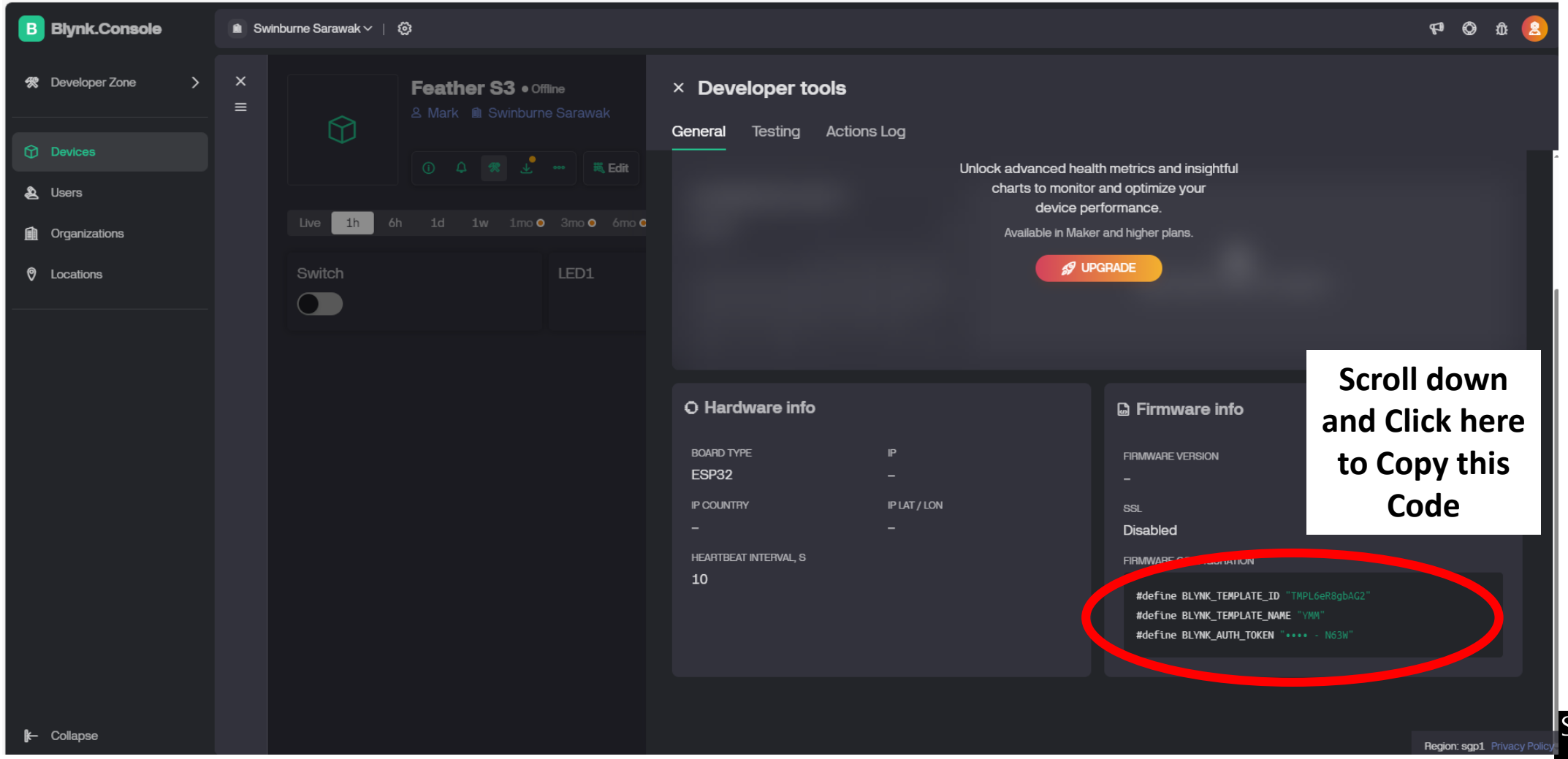
- Once created, click on the Device name to go to it's Dashboard.



- We need to copy that Identification Code. To do this, go to the “Developer Tools” in your Device’s Dashboard.



- In the Developer Tools panel, scroll down until you see the “FIRMWARE CONFIGURATION” section.
- Left-click it once to copy the Identification Code. We will need to paste this into our Arduino Sketch.



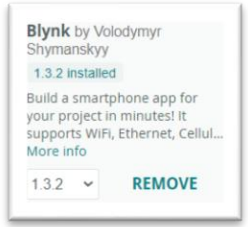
Then, Return to Arduino IDE and NodeMCU ESP32

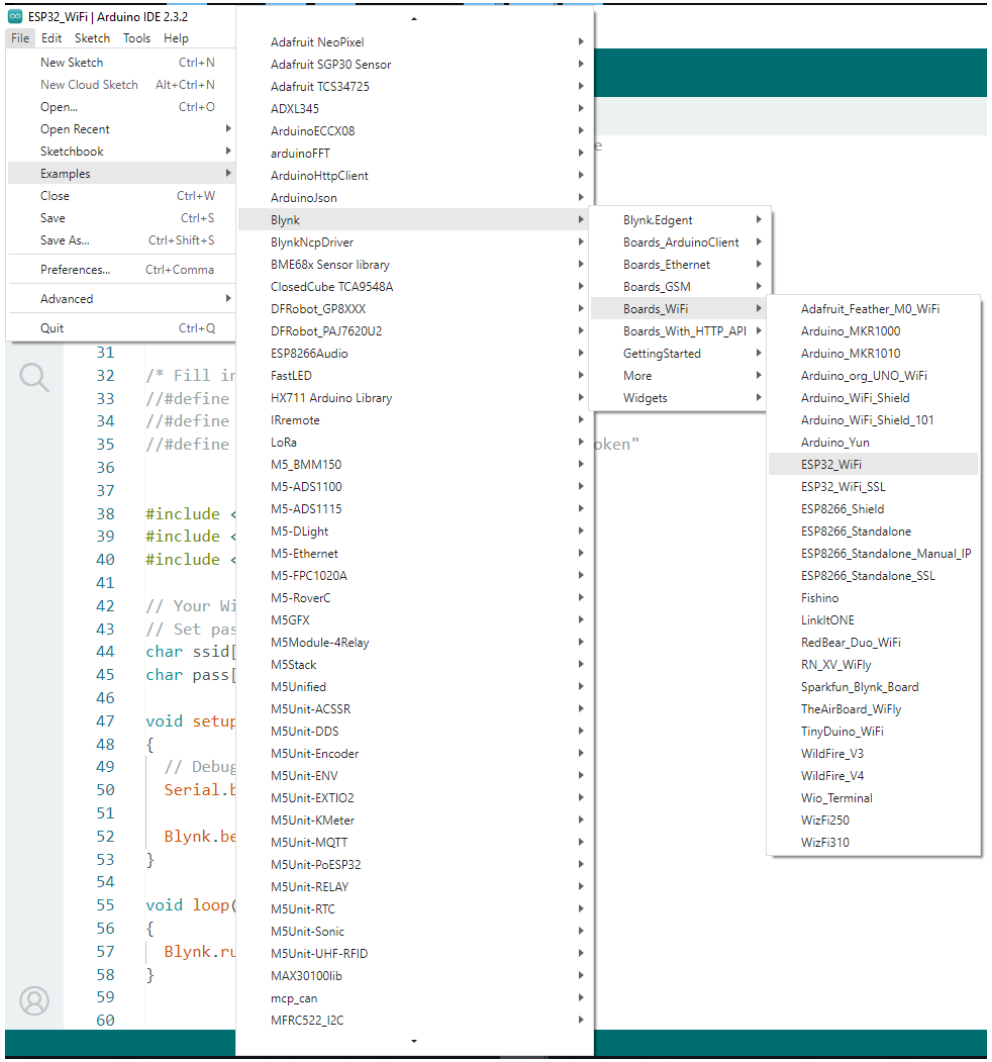
- Connect your NodeMCU ESP32 to the PC and configure your Arduino IDE to prepare programming it.
- Go to Sketch > Include Library > Manage Libraries… panel.
- Search “Blynk”. Install the latest version of Blynk by Volodymyr Shymanskyy.
- Go to File > Examples > Blynk > Boards_WIFI > ESP32_WIFI
- Save a new copy of this Sketch for your machine. (Don’t overwrite the example file).
- Paste the Identification Code from your Blynk into this part of the Sketch:

```
/* Fill in information from Blynk Device Info here */
//#define BLYNK_TEMPLATE_ID           "TMPxxxxxx"
//#define BLYNK_TEMPLATE_NAME         "Device"
//#define BLYNK_AUTH_TOKEN            "YourAuthToken"
```

- Don’t forget to input your WIFI for the ESP32 to connect to:

```
// Your WiFi credentials.
// Set password to "" for open networks.
char ssid[] = "YourNetworkName";
char pass[] = "YourPassword";
```

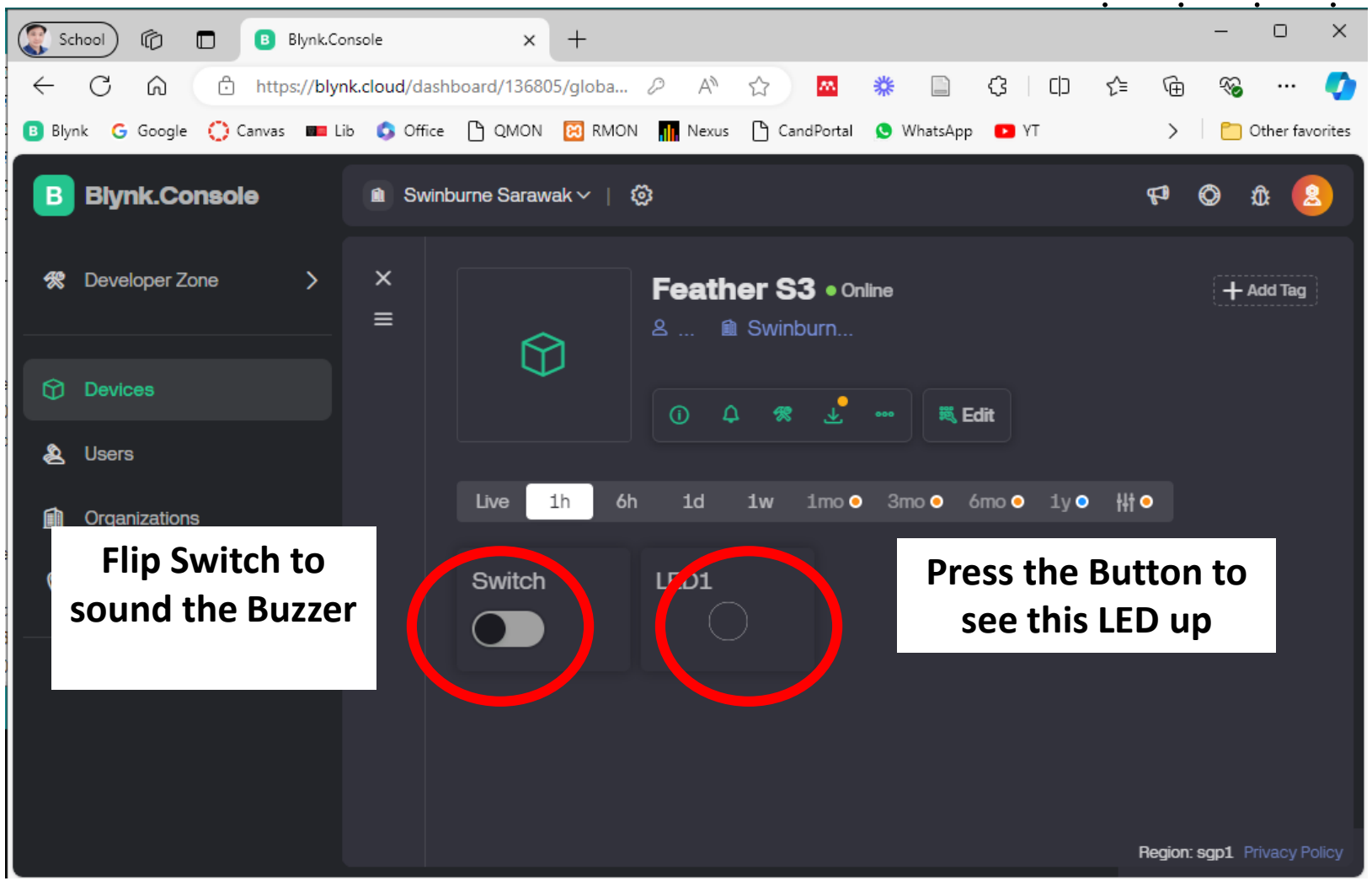




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- Refer to the other examples for getting V0 and V1 to interact with your ESP32.
- Find the example files here:

  File > Examples > Blynk > Getting Started
- The files:
  - GetData – Blynk_Write for updating from Blynk
  - PushData – Blynk TimerEvent for updating to Blynk



Takeaway for this Week

1. Blynk is useful for quick testing and controlling your ESP32 machines (projects wired to the ESP32 board.
2. You can use it for free but functionality is limited.
3. Refer to the latest official tutorial to learn how to use Blynk properly (these get updated quite frequently)

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Presented by Dr. Mark Tee Kit Tsun (SUTS)

If you have questions, find me on MS Teams: mtktsun@swinburne.edu.my